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A stereotetrad-centered approach toward pironetin: Dead ends, Detour, and evolution of the synthetic strategy

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ABSTRACT

Pironetin is a unique α-tubulin inhibitor and has been of utmost interest in the synthetic community. Based on the enabling method to access various stereotriads and stereotetrads, we devised a new total synthesis of pironetin, a total 13 steps from commercially available (*S*)-Roche ester, representing one of the shortest synthetic routes reported to date. The facile fragmentation induced by the deprotonation of the pyrone ring at C4 led us to reorient the installation of the remote enone moiety prior to the lactone ring formation. Moreover, the undesired Michael addition of a phosphate reagent to the acrylate guided us to introduce a sterically demanding cinnamate motif that was also found to facilitate the selective removal of the TBDPS group. The evolution of the synthetic route underlines that the experimental execution of the reactivity of various functional groups is instrumental to devise a successful synthesis.
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1. Introduction

Anticancer agents targeting microtubules are among the most powerful drugs used for chemotherapy cancer treatments [1]. All such therapeutics in current clinical use were designed to bind β-tubulin, a subunit that forms an α/β-heterodimer that is assembled into a filamentous tube-shape structure and plays a pivotal role in cell division during various stages of mitosis [2]. However, during chemotherapy, the overexpression of β-tubulin isoforms that may bind less efficiently to anticancer drugs induces severe drug-resistance [3]. While the quest of more efficient β-tubulin inhibitors is ongoing, an alternative approach for drug development is to evaluate new leads that selectively interact with α-tubulin. Pironetin is the only natural product currently known to bind to α-tubulin [4]. The Chen and Steinmetz groups independently characterized the α-tubulin-pironetin complex and revealed that pironetin covalently binds to Cys-316 of α-tubulin via a Michael addition (Fig. 1) [5,6]. The crystal structures further dictate that pironetin perturbs the T7 loop and helix H8 of α-tubulin that are crucial for establishing longitudinal tubulin contacts in microtubules [6]. This structural information provides a solid molecular basis for understanding the mechanism of the inhibition of the formation of microtubules. Thus, pironetin is considered to be a promising lead compound for the inspiration to develop a new generation of analogues with desirable properties.

Pironetin features a stereotetrad polypropionate chain and a 4,5-disubstituted pyrone motif and was independently isolated from the fermentation broths of *Streptomyces* species by two research groups [7–9]. In addition to its potential use as a potent inhibitor of the cell cycle progression at the M-phase (IC_{50} 5–25 ng/mL) [10], its potency as a plant growth regulator may be beneficial for producing dwarf plants [8b]. Pironetin also exhibits potent immunosuppressive activity on the responses of T and B lymphocytes to mitogens [9]. Due to its intriguing bioactivities and relatively low molecular weight compared to the other known tubulin regulators, pironetin has been of utmost interest for the synthetic community. Since the first total synthesis and structural confirmation of pironetin (PA-48153C) by Kawada and coworkers in 1995 [11], twelve synthetic routes and several structural derivatizations of pironetin have been reported [12–14]. The conventional synthetic strategies largely relied on the classical enantioselective aldol reactions and allylations to construct the stereotetrad fragment (C7–C10). Despite numerous synthetic efforts toward such structurally significant polypropionates in the past decades [14], a practical and versatile synthetic route to this key subunit and

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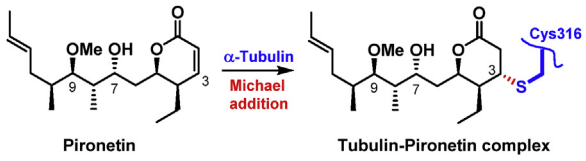


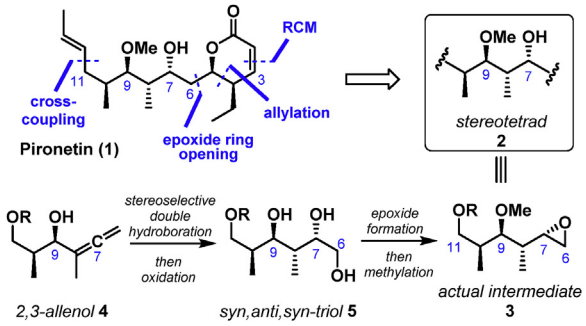
Fig. 1. Mode of action (MoA) of pironetin to α-tubulin.

further application for accessing various analogues remain insufficient. During our effort to develop a non-aldol approach to construct the polypropionate subunit from readily available materials [15], a highly stereoselective double hydroboration of 2,3-allenols was developed [16], thus providing an alternative to the aldol approach for accessing various stereotriads and stereotetrads [17] in a concise manner. To demonstrate the utility of this intriguing method, we implemented it into the total synthesis of pironetin and described the evolution of the synthetic strategy by adjusting the order of reaction arrays for practicability.

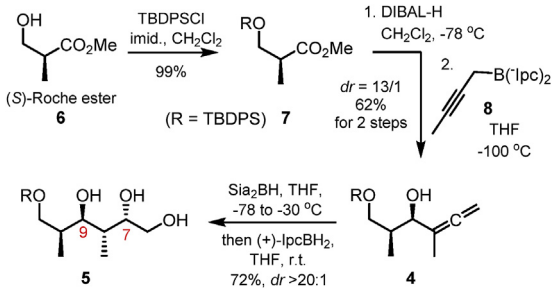
2. Results and discussion

To evaluate a modular strategy for future derivatization, we envisioned the key C7–C10 fragment bearing the four contiguous stereogenic centers (*stereotetrad*) as an essential module to be established with the appropriate functional groups for the subsequent chain elongation. Accordingly, the left alkene chain and the right pyrone motif could be readily implemented and modified during the synthesis that features allylation, ring-closing metathesis, and cross-coupling reactions. With this strategy in mind, the synthetic plan is summarized in Scheme 1. As an equivalent to stereotetrad **2**, the primary alcohol at C11 and epoxide at C6 in intermediate **3** are therefore reserved for further conjunction. The key intermediate epoxide **3** can be derived by a regioselective epoxide formation and concise methylation from triol **5** which is prepared from a stereoselective double hydroboration of 2,3-allenol **4** [16].

Preparation of epoxide 3. The synthesis commenced with the commercially available (*S*)-Roche ester **6** (Scheme 2). After the protection of the primary alcohol by a sterically bulky tert-butyldiphenylsilyl (TBDPS) group, the corresponding methyl ester **7** was subjected to reduction with diisobutylaluminum hydride (DIBAL-H) and was immediately applied to alkynylation with an in situ prepared Brown propargylborane **8** [18]. The diastereoselectivity of this addition was determined to be 13/1 and the major *syn*-2,3-allenol **4** was obtained in a combined 62% yield after two steps. The preparation of compound **4** was readily scaled up to the decagram scale and the selectivity remained excellent. Accordingly, under the optimal conditions as reported in our



Scheme 1. Initial synthetic plan and the key method for accessing stereotetrad **3** from 2,3-allenol **4**. (R = TBDPS).

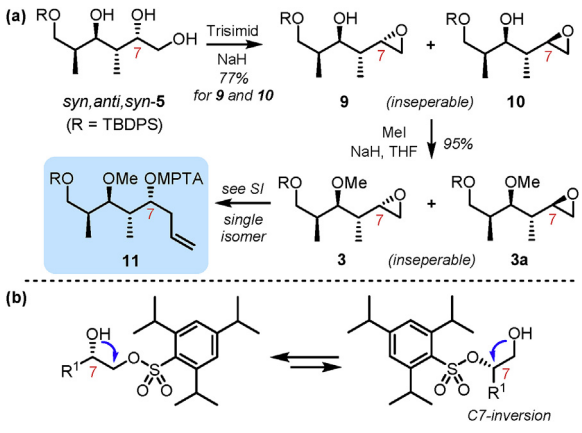


Scheme 2. Preparation of *syn, anti, syn*-triol **5**.

previous study [16], the next double hydroboration was achieved with an excellent diastereoselectivity (*ds* 20/1).

Favorable *syn*, *anti*, *syn*-triol **5** was obtained on gram scale in 72% yield. The subsequent formation of the terminal epoxide proved to be challenging due to the mutual interference of the three proximal hydroxy groups.

At the outset, the Forsyth protocol [19] was intuitively adopted due to its generally good selectivity. The sterically bulky 2,4,6-triisopropylphenylsulfonyl group was rationalized to react with the primary alcohol, followed by an intramolecular S_N2 substitution via the adjacent secondary alcohol to form the requisite epoxide. However, application to triol **5** resulted in two *hard-to-separate* cyclized products in a ratio of 2/1 (**9** and **10**) (Scheme 3a). The mixed epoxides were subjected to the methylation and ring-opening reaction by allylation, the isolable major diastereoisomer was further converted into the Mosher ester **11** that was confirmed to be the desired (*R*)-C7-isomer [20]. To our surprise, the minor cyclized product **10** was identified as a C7-epimer of compound **9**. This unprecedented inversion during the ring-forming process is presumably derived from the migration of the sulfonyl group from the primary alcohol to the secondary alcohol under basic conditions, thus leading to a concomitant formation of epoxide **10** (Scheme 3b). A general survey of different sulfonyl reagents was then undertaken to improve the yield and selectivity (entries 2–5, Table 1). We noticed that, with usage of other less sterically bulky sulfonyl reagents, a considerable amount of the sulfonylated intermediate was accumulated and the ring-forming product was isolated in low yields (17–36%), indicating that the cyclization step is a rate-determining step for such complex substance **5**. Although KHMDS can improve the overall yield of epoxides **9** and **10**, the selectivity remained unsatisfactory (entry 6 vs 1). After no significant improvement in the yield was obtained by increasing the



Scheme 3. Formation of the epoxides.

Table 1
Optimization of the formation of epoxide **9**.

entry	conditions[a]	T (°C)	t (h)	conv.[b]	ratio (9/10)[c]	yield [d]
1	NaH (4 eq), Trisimid, THF[e]	−25	20	100%	2/1	77%
2	NaH (4 eq), TsCl, THF	rt	1.5	66%	4.5/1	17%
3	NaH (4 eq), NsCl, THF	rt	1.5	100%	8/1	18%
4	NaH (4 eq), 4-MeOPhSO$_2$Cl, THF	rt	1.5	100%	–	–
5	NaH (4 eq), TrisCl, THF	rt	1.5	83%	1/2	36%
6	KHMDS (1.3 eq), Trisimid, THF	0	1	100%	1/1	90%
7	NaH (4 eq), Trisimid, THF	−78 to −25	6	100%	2/1	84%
8	DEAD, PPh$_3$, toluene	reflux	4	100%	4/1 [f]	87%
9	DIAD, PPh$_3$, toluene	130	8	100%	12/1 [f]	91% (81%) [g]

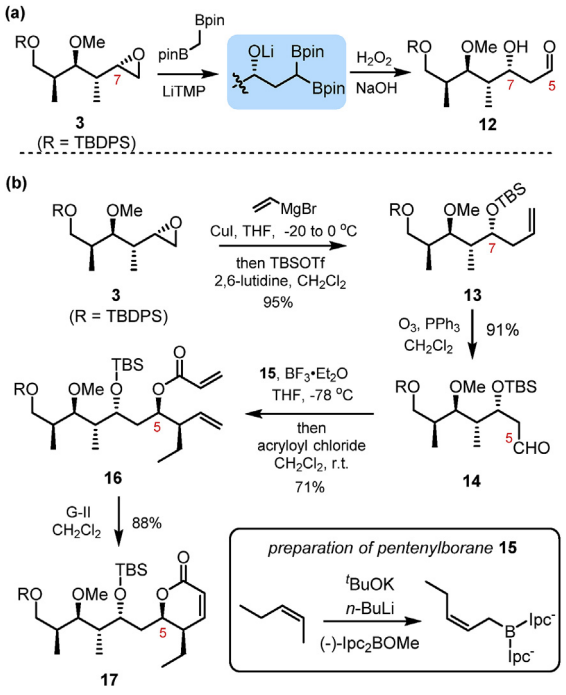
[a] *Syn*, *anti*, *syn*-triol **5** was used on 30-mg scale for optimization.
[b] The conversion of triol **5** was determined by ^{1}H NMR (400 MHz, CDCl$_3$).
[c] The ratio of three cyclized products was determined by ^{1}H NMR (400 MHz, CDCl$_3$).
[d] The combined yield of **9** and **10** was determined by ^{1}H NMR (400 MHz, CDCl$_3$) with CH$_2$Br$_2$ as the internal standard.
[e] The reaction was carried out on 400-mg scale.
[f] A five-membered side product was found (~6% yield) and its structural determination was described in ref. 16.
[g] The isolated yield of **9** was in the parenthesis.

equivalents of the sulfonyl reagent (data not shown) and varying the reaction temperature (entry 7 vs 1), we turned to an approach using the Mitsunobu reaction conditions (entries 8 and 9) [12c,21]. An adduct of phosphine with azocarboxylate is known to form a zwitterionic intermediate that serves as a base to activate the nucleophile. Meanwhile, the resulting ion pair also activates the alcohol by forming a good leaving group [22]. To diminish the side pathway of 5-*exo*-*tet* cyclization, external base-free conditions in toluene with diisopropyl azodiformate (DIAD) were effective to generate epoxides **9**/**10** in a combined 91% yield favoring the desired **9** (*dr* 12/1) (entry 9). The reaction can be readily scaled up to gram scale and excellent selectivity was retained. Subsequently, methylation with methyl iodide proceeded smoothly to afford **3** in excellent yield. A five-membered tetrahydrofuran (6% yield) was also generated due to an involvement of the distal hydroxy group at C9 via the 5-*exo*-*tet* cyclization pathway [16].

The reactivity of pyrone. With the key module **3** in hand, we sought to introduce the C6-aldehyde moiety via a one-carbon elongation inspired by the recent reports by Meek and others [23,24]. Literally, a direct procedure is proposed to realize the epoxide ring-opening by a lithiated di-[B (pin)$_2$]-methane reagent and the subsequent formation of γ-hydroxy bisboronate [23] that can be oxidatively deboronated to form β-hydroxy aldehyde **12** (Scheme 4a). However, this approach received no success after several attempts by varying the reagents, reaction temperature, and solvents. Therefore, we switched to a three-step protocol including vinylation [24], silylation, and subsequent ozonolysis to complete the requisite moiety (Scheme 4b).

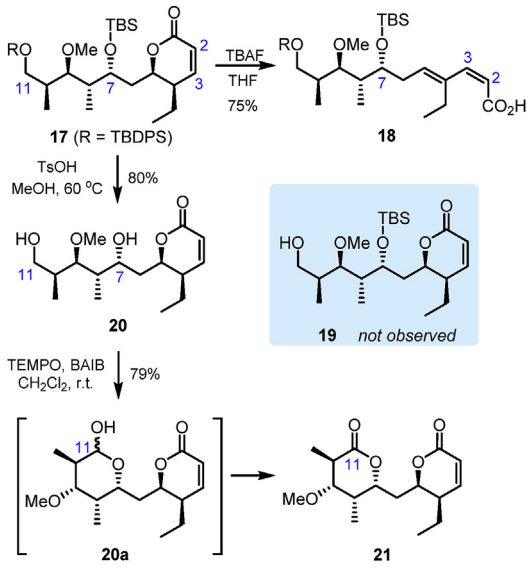
It was interesting to find the first two reactions could be combined without isolation of the intermediate allylic alcohol. After subsequent ozonolysis of alkene **13**, the resulting aldehyde **14**, as a surrogate as the TBS-protected form of aldehyde **12**, was subjected to subsequent allylation. Hatakeyama and coworkers [25] extended the Brown allylation method to the pentenylation of aldehyde in which the *Z*-pentenyl diisopinocampheylborane **15** was prepared in situ by Schlosser's method [26]. This tactic was also independently developed by the Cossy group [27] in the total synthesis of pironetin and other polyketides bearing such a unique pyrone subunit [28]. The *syn*-allylation of aldehyde **14** was then carried out and it was subsequently trapped by acryloyl chloride in a one-pot procedure to deliver the acryl ester derivative of *syn*-stereoisomer **16** in 71% yield with a ratio of 13/1. The following RCM reaction by the Grubbs-II catalyst [12i,29] smoothly produced pyrone **17** in 88% isolated yield.

At this stage, we were optimistic that the planned construction of the left chain (C12−C14) would be straightforward. To our

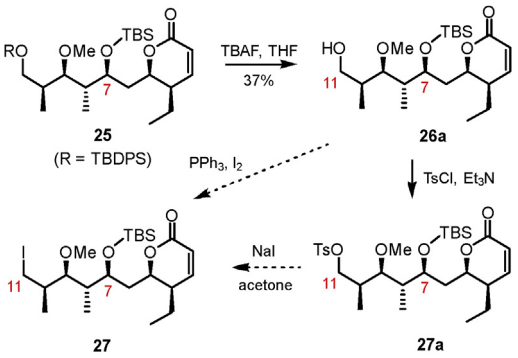


Scheme 4. Preparation of pyrone **17**. (a) The original design; (b) Alternative route for the chain elongation.

surprise, treatment of pyrone **17** with tetrabutylammonium fluoride (TBAF) [30] only resulted in a dienoic acid **18** instead of releasing the mono-desilylated compound **19** bearing a primary alcohol at C11 (Scheme 5). The compound **18** was apparently formed when the allylic proton at C4 was removed under basic conditions, via an internal (*E*1cb-type) elimination way [31]. Acidic conditions enforced with *para*-toluenesulfonic acid (TsOH) in MeOH [12g] removed the two silyl groups to afford diol **20** and no desired **19** was detected. The experimental result further convinced us that this synthetic design would be a dead end when the next oxidation only gave bislactone **21** via a facile generation of intermediate lactol **20a**. The challenges for the elimination and nonselective silyl removal (C7/C11) led us to investigate the origin of chemical selectivity through a re-evaluation of the C7-epimer **3a** (Scheme 6). Following the same reaction sequence, vinylation, silylation, and ozonolysis were applied and aldehyde **23** was obtained in excellent yield. Subsequent allylation and ring closing



Scheme 5. Deprotection of the TBDPS group.

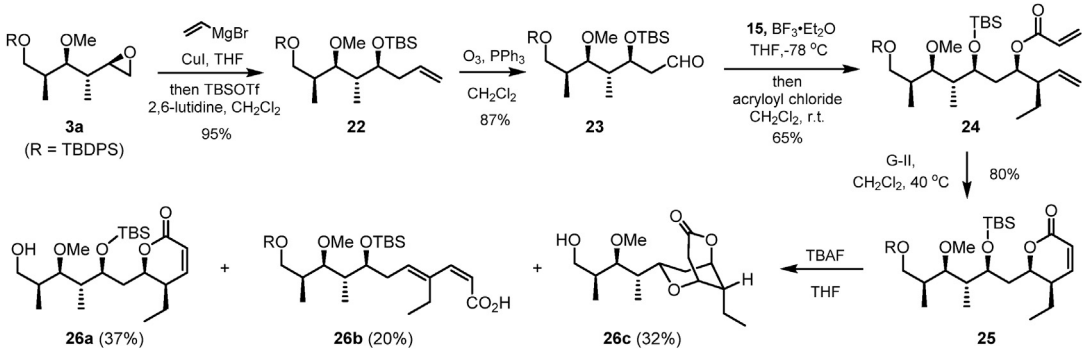


Scheme 7. Attempted Reactions at C11.

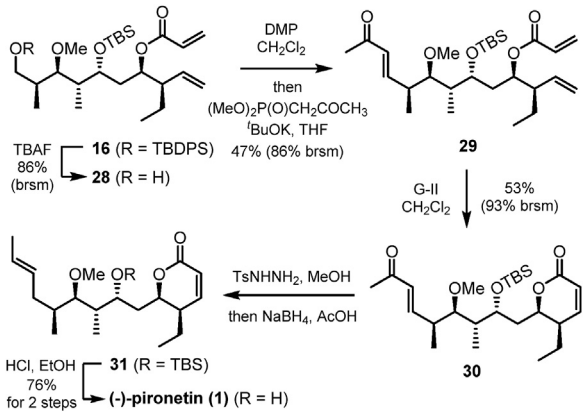
metathesis were carried out to produce pyrone **25**. Interestingly, in the presence of TBAF, although the removal of the TBDPS group was still unsatisfactory, the desired alcohol **26a** was isolated in 37% yield along with dienic acid **26b** (20%) and caged compound **26c** in 32% yield. The latter is attributed to oxa-Michael cyclization with a free hydroxy group at C7 while no further oxa-Michael cyclization occurred to the diol **20**, indicating the thermodynamic control for the bicyclic tetrahydro-α-pyrone skeleton as shown in other bioactive polyketides [32]. The slight selectivity (~2/1) favoring the selective removal of the TBDPS group implies a possible shielding effect due to the proximity of the pyrone ring to the OTBS group at C7 in compound **25**.

Reordering the reaction arrays. To evaluate a possible cross-coupling tactic as proposed in the initial plan (Scheme 1), we utilized alcohol **26a** to prepare the desired fragment (Scheme 7). However, neither $Ph_3P{-}I_2$ [33] nor a two-step protocol (TsCl, triethylamine, then NaI in refluxed acetone) [34] produced the desired iodination product **27**. Although tosylation occurred to yield **27a**, the subsequent cross-coupling using various transition metal-catalyzed conditions [35] failed to deliver the requisite C12–C14 vinylic motif. The low propensity toward nucleophilic substitution of the tosylate derivative **27a** may be attributed to the steric hindrance surrounding the reaction site, which is in sharp contrast to the smooth transformation of the truncated structure in Cossy's synthesis of pironetin.[12i]

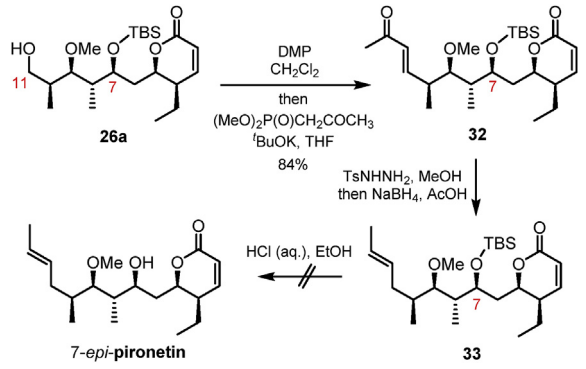
The different reactivities of **17** and **25** toward desilylation conditions provide insight into a possible conformational impact rooted in the adjustment of the single stereogenic center. The fragmentation of the pyrone ring suggests that the ring forming step via the RCM reaction should be resumed at the late stage of the synthetic sequence. Therefore, installation of the left chain was undertaken prior to the RCM reaction for the generation of the lactone. At this stage, Kawada's elegant solution [11] of introducing the vinylic moiety via a sigmatropic rearrangement of allylic diazene [36] was intriguing because the reaction site is away from the condensed stereotetrad moiety (C7–C10). Accordingly, compound **16** was subjected to desilylation with TBAF and the primary alcohol **28** was obtained in 86% yield (brsm) (Scheme 8). Subsequent oxidation by Dess-Martin periodinane (DMP) [37] and the Horner–Wadsworth–Emmons (HWE) reaction [38] were performed following a one-pot protocol to deliver the requisite *E*-enone **29** in 47% yield (Scheme 8). It should be noted that a severe side reaction occurred via the Michael addition of the phosphate to the acrylate motif at the right side of the molecule and thus the desired product **29** was then gradually consumed [39]. A good combined yield (80% after two cycles) was obtained if the recovered starting material was reintroduced into the reaction system. For the feasibility of the current strategy toward the final target, following Kawada's pioneering work, a convenient hydrazine-reduction-rearrangement cascade was adapted to smoothly produce **31** in 85% yield. Interestingly, we also applied the same reaction sequence to the 7-epimer **26a** and the complete carbon framework (i.e. **32** and **33**) was established. However, an acid removal of the TBS group of **33** resulted in several unidentified products (Scheme 9). In striking contrast, the acidic conditions allowed the removal of the TBS group in **31** and the target molecule was obtained (Scheme 8). We are delighted that the ketone reduction and desilylation could be



Scheme 6. A challenge in the removal of the TBDPS group.



Scheme 8. Completion of the synthesis of pironetin (**1**).

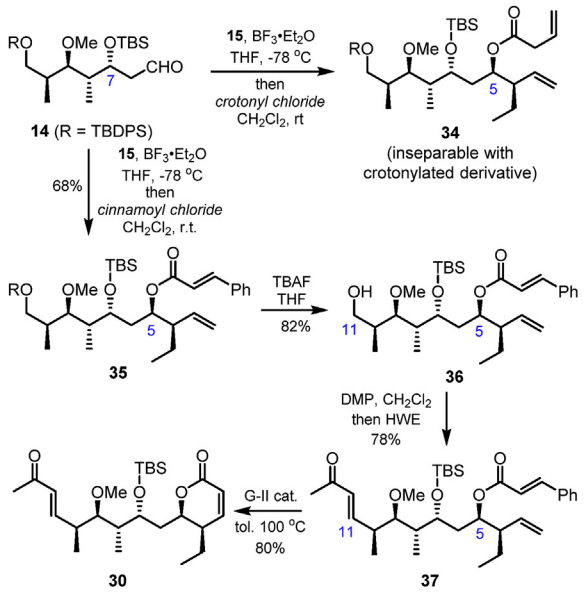


Scheme 9. Attempted synthesis of 7-*epi*-pironetin.

performed following a one-pot protocol to deliver pironetin (**1**) in 76% overall yield. The synthetic sample is completely identical with the reported data [7,8].

Evolution of a synthetic strategy. The aforementioned synthetic sequence has two critical shortcomings that should be ameliorated: (1) to diminish the undesired removal of the *tert*-butyldimethylsilyl (TBS) group, delicate control of the reaction condition must be used that may be not reproducible on a larger scale; (2) the severe Michael addition to the acrylate group [39] occurring before or after the HWE reaction reduces the isolated yield. Consequently, the recycling of the starting materials for the above two transformations is tedious and inefficient. Therefore, a modification of the synthetic strategy is urgently needed. As we observed, the adjacent sterically demanding groups such as the pyrone ring and acrylate group attenuate the reactivity of C7-OTBS during the deprotection of TBDPS group at C11. We envisioned that introducing an additional substituent on the acryl group, such as a crotyl or cinnamyl group, may be helpful for shielding the C7-silyloxy group under the desilylation conditions, and more importantly, the extent of the Michael addition reaction to the α,β-unsaturated ester can be diminished due to the steric hindrance at the β-position.

Accordingly, the homoallylic alcohol derived from the allylation of aldehyde **14** was immediately quenched by esterification with crotonoyl chloride and cinnamoyl chloride, respectively. However, crotonoylation proved to be inappropriate due to the migration of the double bond (i.e., **34**) in the presence of an organic base [40] (Scheme 10). A preliminary screening of different bases such as triethylamine and diisopropylethylamine did not efficiently prevent such an isomerization. On the other hand, in the presence of



Scheme 10. The improved synthesis strategy.

triethylamine, cinnamate **35** was obtained in good overall yield (78%).

To our delight, the next desilylation indeed proceeded well and the primary alcohol **36** was obtained in 82% yield without an appreciable amount of diol (such as a compound like **20** in Scheme 5) contamination. The subsequent oxidation and HWE reaction were carried out smoothly at an elevated temperature (45 °C) and *E*-enone **37** was obtained in 78% isolated yield without the denoted impurity derived from Michael addition. Although the completion of the RCM reaction required an elevated temperature (100 °C) in toluene, the resulting enone-pyrone **30** was obtained in a good isolated yield (80%). Repeating the established one-pot protocol including the enone reduction and desilylation with HCl completed the synthesis of pironetin (**1**).

3. Conclusion

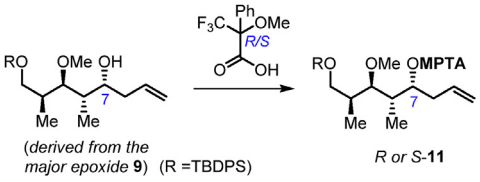
In summary, we discussed here the evolution of a synthetic strategy toward pironetin by changing the order of the reaction arrays based on the dead ends and inefficiency encountered in the synthetic efforts [41]. Several notable features can be drawn from here: (1) the facile fragmentation of the pyrone ring should be taken into consideration in future synthesis because this structural motif is embedded in several bioactive polyketides; (2) the use of the acryl group that commonly used for the construction of the pyrone ring via an RCM reaction involves a Michael addition when a soft nucleophile is applied and the cinnamyl group is suggested to be utilized as a surrogate; (3) a conformational change derived from the one-site switch of the chiral centers is also crucial for an array of the subsequent chemical transformations. These findings are clearly valuable for devising an efficient synthesis for complex polyketides with the criteria of practicability [42], even for a target molecule as small as pironetin. Guided by the versatility of the stereotetrad-centered strategy, application in the construction of more complex bioactive polyketides is currently underway in this laboratory and will be reported in due course.

4. Experimental section

General procedure: All non-aqueous reactions were run under

a positive pressure of argon. Anhydrous solvents were obtained using standard drying techniques. All commercial grade reagents were used without further purification unless stated otherwise. Flash chromatography was performed on 300–400 mesh silica gel with the indicated solvent systems. ^{1}H NMR spectra were recorded on Bruker 400 (400 MHz) spectrometer or Agilent DD2 (500 MHz) and chemical shifts are reported in ppm down field from TMS, using TMS (0.00 ppm) or residual chloroform (7.26 ppm) as an internal standard. Data are reported as: (s = singlet, br = broad, d = doublet, t = triplet, q = quartet, hept = heptalet, m = multiplet; J = coupling constant in Hz, integration.). ^{13}C NMR spectra were recorded on Bruker 400 (100 MHz) spectrometer or Agilent DD2 (125 MHz), using proton decoupling unless otherwise noted. The optical rotation was recorded on a JASCO P-1030 polarimeter. $[\alpha]_D$ values were given in 10^{-1}**.** deg**.**cm^2**.** g^{-1}. HRMS were recorded by using either FTMS-7 or IonSpec 4.7 spectrometers. FT-IR spectra were recorded on Nicolet iN 10 MX. Frequencies are given in reciprocal centimeters (cm^{-1}), and only selected absorbance is reported.

The experimental details about the preparation and characterization data of compounds **3**, **4**, **6**, **7**, **9**, **13**, **14**, **16**, **28** and **29** were recorded in the previous work [16]. The detailed structural determination of epoxide **9** by the Mosher method [43] as the MPTA derivatives **11** is described in the online Supporting Information. The experimental procedure is described in the following section.



N, *N*-Dimethylformamide (2.5 μL, 0.03 mmol) was added to (*R*)-2-methoxy-2-phenyl- 3,3,3-trifluoropropanoic acid or (*S*)-2-methoxy-2-phenyl-3,3,3-trifluoropropanoic acid (17 mg, 0.073 mmol) and oxalyl chloride (31 μL, 0.36 mmol) in hexane (0.7 mL), and the solution was stirred for 2 h. The mixture was filtered and the filtrate was concentrated under reduced pressure to afford the corresponding acid chloride as colorless oil. The crude acid chloride was re-dissolved in CH_2Cl_2 (1.0 mL) and then added into a solution of alcohol (*derived from the major epoxide* **9**) (6.4 mg, 0.014 mmol), triethylamine (21 μL, 0.14 mmol) and DMAP (1.0 mg) in CH_2Cl_2 (1.0 mL). The reaction mixture was stirred for 6 h. Ether (5 mL) was added and the solution was washed with water and brine, and then dried over Na_2SO_4. The mixture was filtered, and the filtrate was concentrated under reduced pressure. The residue was subjected to purification on PTLC (PE/EA = 30/1) to give the (*R*)-MPTA-**11** (5.2 mg, 55% yield) ester or (*S*)-MPTA-**11** (6.0 mg, 63% yield) as colorless oil.

(4R,5S,6R,7S)-8-((tert-butyldiphenylsilyl)oxy)-6-methoxy-5,7-dimethyloct-1-en-4-yl-*(R)-3,3,3-trifluoro-2-methoxy-2-phenylpropanoate* (***R*-11**) ^{1}H NMR (400 MHz, $CDCl_3$): 7.69–7.62 (m, 4H), 7.59–7.53 (m, 2H), 7.46–7.31 (m, 9H), 5.70 (ddt, J = 17.2, 10.1, 7.1 Hz, 1H), 5.53 (dd, J = 7.7, 6.4 Hz, 1H), 5.04 (dd, J = 20.3, 4.9 Hz, 2H), 3.65–3.58 (m, 1H), 3.56–3.47 (m, 4H), 3.39 (s, 3H), 3.21 (dd, J = 9.9, 1.8 Hz, 1H), 2.49 (dt, J = 13.9, 6.9 Hz, 1H), 2.28 (dt, J = 14.3, 7.3 Hz, 1H), 1.91–1.74 (m, 2H), 1.05 (s, 9H), 0.84 (d, J = 7.0 Hz, 3H), 0.73 (d, J = 6.9 Hz, 3H). ^{13}C NMR (150 MHz, $CDCl_3$): 166.29, 135.72, 135.69, 133.93, 133.46, 132.11, 129.82, 129.79, 129.64, 128.45, 127.85, 127.82, 127.80, 123.53 (q, J = 288.5 Hz), 118.34, 81.01, 75.75, 66.79, 61.20, 55.53, 38.36, 37.99, 37.26, 27.01, 19.37, 10.07, 9.43.^{19}F NMR (565 MHz, $CDCl_3$): −71.36 (s). HRMS-ESI Positive: calcd. for $C_{37}H_{47}O_5F_3NaSi$ $[M+Na]^+$: 679.3037; found: 679.3042.

(4R,5S,6R,7S)-8-((tert-butyldiphenylsilyl)oxy)-6-methoxy-5,7-dimethyloct-1-en-4-yl-*(S)-3,3,3-trifluoro-2-methoxy-2-phenylpropanoate* (***S*-11**) ^{1}H NMR (400 MHz, $CDCl_3$): 7.65 (dd, J = 7.2, 6.0, 1.5 Hz, 4H), 7.58 (dd, J = 6.4, 2.8 Hz, 2H), 7.46–7.33 (m, 9H), 5.75 (ddt, J = 17.1, 10.1, 7.1 Hz, 1H), 5.56–5.50 (m, 1H), 5.14–5.04 (m, 2H), 3.63–3.56 (m, 1H), 3.55–3.47 (m, 4H), 3.38 (s, 3H), 3.19 (dd, J = 10.0, 1.8 Hz, 1H), 2.61 (dt, J = 13.5, 6.6 Hz, 1H), 2.44–2.31 (m, 1H), 1.89–1.73 (m, 2H), 1.04 (s, 9H), 0.82 (d, J = 7.0 Hz, 3H), 0.70 (d, J = 6.9 Hz, 3H). ^{13}C NMR (150 MHz, $CDCl_3$): 166.20, 135.73, 135.71, 133.95, 133.52, 132.36, 129.77, 129.75, 129.69, 128.47, 127.79, 127.77, 127.71, 123.63 (q, J = 288.5 Hz), 118.36, 80.73, 76.02, 66.74, 61.14, 55.42, 38.26, 37.89, 37.17, 26.95, 19.32, 9.72, 9.31.^{19}F NMR (565 MHz, $CDCl_3$): −70.68 (s). HRMS-ESI Positive: calcd. for $C_{37}H_{47}O_5F_3NaSi$ $[M+Na]^+$: 679.3037; found: 679.3039.

(2S,3R,4R,5S)-6-((tert-butyldiphenylsilyl)oxy)-3,5-dimethylhexane-1,2,4-triol (***5***).

Preparation of Sia_2BH: To a solution of 2-methyl-2-butene (2.5 mL, 23.1 mmol) in THF (9.3 mL) was added dropwise $BH_3 \cdot DMS$ (1.06 mL, 10.5 mmol) at 0 °C. The resulting mixture was stirred at 0 °C for 2 h, and then a solution of Sia_2BH in THF was prepared.

Preparation of $(+)$-$IpcBH_2$: To a solution of $((+)\text{-IpcBH}_2)_2 \cdot$TMEDA (2.55 g, 6.0 mmol) in THF (12 mL) was added dropwise $BF_3 \cdot Et_2O$ (1.45 mL, 11.7 mmol) at room temperature. The resulting mixture was stirred at room temperature for 2.5 h and then the resulting slurry was filtered into a flask, and then a solution of $(+)$-$IpcBH_2$ (~3.5 eq) in THF was prepared.

To a solution of allene **4** (3.0 mmol) in THF (6.0 mL) was added dropwise the aforementioned solution of Sia_2BH at −78 °C. After 30 min, then resulting mixture was warmed up to −50 °C. After 1 h, the reaction mixture was warmed up to −30 °C and stirred for 1 h. The above solution of $(+)$-$IpcBH_2$ was added and warmed up to room temperature and stirred for 5 h. The reaction mixture was quenched by addition of MeOH (0.9 mL) at 0 °C (*Caution: gas evolution*), followed by slow addition of NaOH (aq. 3 N, 12.0 mL) and H_2O_2 (30% in H_2O, 12.0 mL), respectively. The resulting mixture was warmed up to room temperature and stirred for another 3 h. Ethyl acetate (60 mL) was added and the mixture was filtered by Celite®. The filtrate was collected and the organic layer was separated and then the aqueous layer was extracted with ethyl acetate (3 × 90 mL). The combined organic extracts were washed by water and brine, and then dried over Na_2SO_4. The solvent was removed under vacuum. Purification by flash column chromatography using silica gel (PE/EA/EtOH: 20/1/1–10/1/1) afforded triol **5** (891 mg, 72% yield) as colorless oil. $[\alpha]_D^{25}$ +16.7 (c = 0.90, $CHCl_3$). ^{1}H NMR (400 MHz, CD_3OD): 7.67 (dd, J = 4.2, 1.6 Hz, 4H), 7.40 (dd, J = 14.6, 7.3 Hz, 6H), 4.01 (t, J = 5.2 Hz, 1H), 3.79 (d, J = 6.6 Hz, 1H), 3.73 (dd, J = 9.8, 6.6 Hz, 1H), 3.63–3.53 (m, 2H), 3.48 (dd, J = 11.0, 5.2 Hz, 1H), 1.88–1.79 (m, 1H), 1.78–1.69 (m, 1H), 1.05 (s, 9H), 0.87 (d, J = 6.8 Hz, 3H), 0.80 (d, J = 7.0 Hz, 3H). ^{13}C NMR (100 MHz, CD_3OD): 136.70, 134.92, 130.83, 128.76, 74.08, 73.02, 68.56, 65.53, 38.85, 38.56, 27.41, 20.05, 10.74, 9.96. FT-IR: ν (cm^{-1}) 3385, 3071, 2962, 2930, 2857, 1472, 1428, 1261, 1112, 1020, 823, 702. HRMS-ESI Positive: calcd. for $C_{24}H_{37}O_4Si$ $[M+H]^+$: 417.2456; found: 417.2453.

(5R,6R)-6-((2R,3R,4R,5S)-2-((tert-butyldimethylsilyl)oxy)-6-((tert-butyldiphenylsilyl)oxy)-4-methoxy-3,5-dimethylhexyl)-5-ethyl-5,6-dihydro-2H-pyran-2-one (***17***). A solution of ester **16** (93 mg, 0.136 mmol) in freshly distilled CH_2Cl_2 (27 mL) was stirred at 40 °C and slowly added a solution of Grubbs-II catalyst (23 mg, 0.027 mmol) in CH_2Cl_2 (1.0 mL). The reaction process was monitored by TLC. After completion, the solvent was removed under reduced pressure. Further purification of the brown residue by flash column chromatography on silica gel (PE/EA: 15:1) afforded pyrone **17** (78 mg, 88% yield) as colorless oil. $[\alpha]_D^{25}$ -55.1 (c = 0.74, $CHCl_3$). ^{1}H NMR (400 MHz, $CDCl_3$): 7.70–7.63 (m, 4H), 7.45–7.35 (m, 6H), 7.00 (dd, J = 9.8, 6.1 Hz, 1H), 6.02 (d, J = 9.8 Hz, 1H), 4.53–4.46 (m, 1H),

4.19 (t, $J = 6.5$ Hz, 1H), 3.64 (t, $J = 9.6$ Hz, 1H), 3.57–3.47 (m, 5H), 2.18 (td, $J = 9.8$, 4.4 Hz, 1H), 2.03–1.93 (m, 1H), 1.94–1.84 (m, 1H), 1.83–1.75 (m, 1H), 1.74–1.64 (m, 1H), 1.59–1.44 (m, 2H), 1.06 (s, 9H), 0.96 (t, $J = 7.5$ Hz, 3H), 0.89 (s, 9H), 0.80 (d, $J = 7.0$ Hz, 3H), 0.69 (d, $J = 6.9$ Hz, 3H), 0.12 (s, 3H), 0.10 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$): 164.62, 150.85, 135.72, 135.70, 133.98, 133.95, 129.75, 129.72, 127.79, 127.76, 120.95, 80.67, 77.51, 68.52, 66.91, 60.89, 41.19, 38.29, 37.95, 37.32, 27.06, 26.19, 20.93, 19.39, 18.48, 11.09, 10.40, 9.53, −3.46, −3.53. FT-IR: ν (cm^{-1}) 3071, 3049, 2959, 2931, 2857, 1730, 1472, 1463, 1428, 1388, 1253, 1112, 1073. HRMS-ESI Positive: calcd. for $C_{38}H_{61}O_5Si_2$ $[M+H]^+$: 653.4052; found: 653.4051.

*(2Z,4E,7R,8R,9R,10S)-7-((tert-butyldimethylsilyl)oxy)-11-((tert-butyldiphenylsilyl)oxy)-4-ethyl-9-methoxy-8,10-dimethylundeca-2,4-dienoic acid (**18**).* To an oven-dried flask charged with lactone **17** (25.2 mg, 0.038 mmol) in anhydrous THF (1.5 mL) was added tetrabutylammonium fluoride (77 μL, 1.0 M in THF). The resulting solution was allowed to stir at room temperature for 2 h, before quenched by aqueous solution of NH_4Cl (2 mL). The aqueous layer was extracted with EtOAc (3 × 10 mL). The combined organic layers were washed with brine, dried over Na_2SO_4, filtered and concentrated. The residue was subjected to purification by flash column chromatography on silica gel (PE/EA: 15/1) to give the dienoic acid **18** (19 mg, 75% yield) as colorless oil. $[\alpha]_D^{25}$ +13.4 ($c = 0.025$, $CHCl_3$). ^{1}H NMR (400 MHz, $CDCl_3$): 7.71–7.61 (m, 4H), 7.46–7.35 (m, 6H), 6.40 (d, $J = 11.5$ Hz, 1H), 5.74 (d, $J = 11.4$ Hz, 1H), 5.39 (dd, $J = 9.8$, 6.3 Hz, 1H), 3.94 (dd, $J = 10.8$, 2.9 Hz, 1H), 3.70–3.57 (m, 5H), 3.50 (dd, $J = 10.3$, 5.6 Hz, 1H), 2.49–2.38 (m, 1H), 2.29–2.09 (m, 3H), 1.75–1.64 (m, 2H), 1.05 (s, 9H), 1.01 (t, $J = 7.6$ Hz, 3H), 0.90 (s, 9H), 0.76 (d, $J = 6.9$ Hz, 3H), 0.63 (d, $J = 6.9$ Hz, 3H), 0.12 (s, 6H). ^{13}C NMR (150 MHz, $CDCl_3$): 169.46, 143.06, 140.05, 135.76, 135.71, 133.88, 129.81, 129.79, 127.84, 127.81, 126.41, 121.56, 82.44, 71.25, 66.38, 61.53, 38.95, 37.75, 34.44, 27.05, 26.09, 23.09, 19.35, 18.31, 13.02, 9.32, 9.09, −3.27, −4.09. FT-IR: ν (cm^{-1}) 2962, 2926, 2855, 1701, 1460, 1427, 1383, 1261, 1017, 800, 702. HRMS-ESI Positive: calcd. for $C_{38}H_{61}O_5Si_2$ $[M+H]^+$: 653.4052; found: 653.4051.

*(5R,6R)-5-ethyl-6-(((2R,3S,4S,5R)-4-methoxy-3,5-dimethyl-6-oxotetrahydro-2H-pyran-2-yl)methyl)-5,6-dihydro-2H-pyran-2-one (**21**).* To a flask charged with pyrone **17** (5 mg, 0.007 mmol) in MeOH (1 mL) was added TsOH (10 mg) in one portion. The resulting solution was heated to 60 °C until the TLC indicated the full consumption of the starting material. The solvent was removed under the reduced pressure and the residue was subjected to purification through a short plug of silica gel. The resulting alcohol was then redissolved in anhydrous CH_2Cl_2 (0.5 mL) and then added TEMPO (2 mg, 0.01 mmol) and (diacetoxyiodo)benzene (6 mg, 0.02 mmol), respectively. The reaction mixture was stirred at room temperature for additional 2 h. The solvent was removed *in vacuo* and the brown organic residue was subjected to flash column chromatography on silica gel (PE: EA = 1/2) to afford lactone **21** (1.8 mg, 79% yield) as colorless oil. $[\alpha]_D^{25}$ -25.7 ($c = 0.17$, $CHCl_3$). ^{1}H NMR (400 MHz, $CDCl_3$): 7.03 (dd, $J = 9.8$, 6.1 Hz, 1H), 6.05 (d, $J = 9.8$ Hz, 1H), 4.83 (dt, $J = 7.9$, 3.6 Hz, 1H), 4.64–4.57 (m, 1H), 3.37 (s, 3H), 3.32 (dd, $J = 9.7$, 4.6 Hz, 1H), 2.51–2.41 (m, 1H), 2.35–2.24 (m, 2H), 1.97–1.82 (m, 2H), 1.73–1.62 (m, 1H), 1.55–1.48 (m, 1H), 1.38 (d, $J = 7.1$ Hz, 3H), 0.99 (t, $J = 7.5$ Hz, 3H), 0.91 (d, $J = 7.1$ Hz, 3H). ^{13}C NMR (100 MHz, $CDCl_3$): 173.47, 164.34, 150.64, 120.90, 82.52, 76.28, 74.84, 56.67, 38.75, 38.52, 35.54, 33.12, 20.79, 14.85, 11.06, 4.70. FT-IR: ν (cm^{-1}) 2962, 2925, 1732, 1456, 1379, 1260, 1200, 1097, 1024, 969, 880, 856, 801. HRMS-ESI Positive: calcd. for $C_{16}H_{25}O_5$ $[M+H]^+$: 297.1697; found: 297.1696.

*(5S,6R,7R,8S)-5-allyl-7-methoxy-2,2,3,3,6,8,12,12-octamethyl-11,11-diphenyl-4,10-dioxa-3,11-disilatridecane (**22**)* To a solution of CuI (35 mg, 0.186 mmol) in anhydrous THF (8.0 mL) was added vinylmagnesium bromide (0.56 M in THF, 8.3 mL) dropwise at −20 °C. The resulting solution was stirred for 30 min at this temperature and then added a solution of epoxide **3a** (384 mg, 0.931 mmol) in THF (6.0 mL). The resulting mixture was then allowed to warm up to 0 °C. The reaction process was monitored by TLC. After completion, the reaction mixture was treated with 2,6-lutidine (1.3 mL, 11.2 mmol) and TBSOTf (1.8 mL, 7.45 mmol). The resulting mixture was further stirred at 0 °C until the alcohol intermediate was completely consumed. The reaction was quenched by slow addition of aqueous solution of NH_4Cl (15 mL). The organic layer was seperated and the aqueous layer was extracted with EtOAc (3 × 10 mL). The combined organic layers were washed with brine, dried over Na_2SO_4, filtered and concentrated *in vacuo*. The residue was subjected to purification by flash chromatography on silica gel (PE/EA: 100/1) to give compound **22** (490 mg, 95% yield) as colorless oil. $[\alpha]_D^{25}$ -11.8 ($c = 2.19$, $CHCl_3$). ^{1}H NMR (400 MHz, $CDCl_3$): 7.70 (dd, $J = 9.4$, 7.8, 1.4 Hz, 4H), 7.48–7.36 (m, 6H), 5.92 (ddt, $J = 17.2$, 10.1, 7.1 Hz, 1H), 5.06 (t, $J = 12.9$ Hz, 2H), 3.98 (dt, $J = 8.9$, 3.0 Hz, 1H), 3.67 (t, $J = 9.6$ Hz, 1H), 3.57 (dd, $J = 9.9$, 6.0 Hz, 1H), 3.45–3.37 (m, 4H), 2.34–2.23 (m, 1H), 2.23–2.11 (m, 1H), 1.94–1.80 (m, 2H), 1.10 (s, 9H), 0.92 (s, 9H), 0.81 (d, $J = 7.0$ Hz, 3H), 0.74 (d, $J = 6.9$ Hz, 3H), 0.07 (s, 3H), 0.06 (s, 3H). ^{13}C NMR (125 MHz, $CDCl_3$): 137.39, 135.73, 134.07, 134.02, 129.76, 127.79, 116.14, 81.57, 73.27, 66.48, 60.72, 42.07, 37.92, 37.08, 27.05, 26.09, 19.43, 18.26, 10.91, 9.32, −4.27. FT-IR: ν (cm^{-1}) 3072, 2958, 2930, 2857, 1472, 1428, 1388, 1255, 1112, 835, 701. HRMS-ESI Positive: calcd. for $C_{33}H_{55}O_3Si_2$ $[M+H]^+$: 555.3684; found: 555.3689.

*(3S,4R,5R,6S)-3-((tert-butyldimethylsilyl)oxy)-7-((tert-butyldiphenylsilyl)oxy)-5-methoxy-4,6-dimethylheptanal (**23**).* Alkene **22** (408 mg, 0.736 mmol) was dissolved in CH_2Cl_2 (16 mL) before cooling down to −78 °C. The solution was treated with ozone until it turned light blue, after which the reaction was flushed with nitrogen gas for 10 min. The reaction mixture was added PPh_3 (0.97 g, 3.68 mmol) in one portion and then stirred for an additional 2 h at room temperature. The solvent was removed under reduced pressure. The residue wa subjected to purification by flash chromatography on silica gel (PE/EA: 60/1) afforded aldehyde **23** (356 mg, 87% yield) as colorless oil. $[\alpha]_D^{25}$ -12.7 ($c = 1.32$, $CHCl_3$). ^{1}H NMR (400 MHz, $CDCl_3$): 9.83 (dd, $J = 2.6$, 1.9 Hz, 1H), 7.70–7.61 (m, 4H), 7.47–7.34 (m, 6H), 4.56 (dt, $J = 9.0$, 3.1 Hz, 1H), 3.63 (t, $J = 9.6$ Hz, 1H), 3.55 (dd, $J = 10.0$, 6.0 Hz, 1H), 3.39 (s, 3H), 3.27 (dd, $J = 9.9$, 1.8 Hz, 1H), 2.58 (dd, $J = 15.6$, 9.1, 2.9 Hz, 1H), 2.49 (dd, $J = 15.6$, 3.0, 1.8 Hz, 1H), 1.93–1.77 (m, 2H), 1.06 (s, 9H), 0.87 (s, 9H), 0.80 (d, $J = 7.0$ Hz, 3H), 0.73 (d, $J = 6.9$ Hz, 3H), 0.06 (s, 3H), 0.05 (s, 3H). ^{13}C NMR (125 MHz, $CDCl_3$): 203.13, 135.71, 135.70, 133.89, 129.82, 127.81, 81.75, 68.59, 66.33, 60.92, 46.69, 42.19, 38.00, 27.04, 25.92, 19.40, 18.12, 11.12, 9.45, −4.23, −4.66. FT-IR: ν (cm^{-1}) 3071, .050, 2930, 2857, 1728, 1472, 1428, 1388, 1258, 1094, 702. HRMS-ESI Positive: calcd. for $C_{32}H_{52}O_4NaSi_2$ $[M+Na]^+$: 579.3296; found:579.3301.

*(3R,4R,6S,7R,8R,9S)-6-((tert-butyldimethylsilyl)oxy)-10-((tert-butyldiphenylsilyl)oxy)-3-ethyl-8-methoxy-7,9-dimethyldec-1-en-4-yl acrylate (**24**).* To a suspension of *cis*-2-pentene (0.46 mL, 4.27 mmol) and potassium *tert*-butoxide (1.4 mL, 1.0 M in THF) in freshly distilled THF (1.5 mL) at −78 °C was slowly added ^{n}BuLi (0.59 mL, 2.4 M in hexanes). The resulting solution was then stirred for 10 min at −50 °C and was re-cooled to −78 °C. After addition of (−)-methoxydiisopinocampheylborane (562 mg, 1.78 mmol) in THF (3.5 mL), the colorless solution was then kept stirring for 30 min at −78 °C and then added boron trifluoride etherate (0.22 mL, 1.78 mmol) and aldehyde **23** (396 mg, 0.71 mmol), respectively. The reaction mixture was stirred for an additional 6 h until the starting material was completely consumed (TLC). The reaction was quenched by slow addition of aqueous solution of NaOH (1.2 mL, 3 N) and 30 wt% H_2O_2 (1.2 mL). The resulting mixture

was heated up and refluxed for 1 h. The aqueous layer was extracted with EtOAc (3 × 10 mL). The combined organic layers were washed with brine, dried over Na_2SO_4, and concentrated under reduced pressure. The residue was used directly for the next step without further purification. To a solution of the above alcohol in CH_2Cl_2 (30 mL) was added triethylamine (1.0 mL, 7.4 mmol) and acryloyl chloride (0.45 mL, 5.5 mmol) at 0 °C. The resulting mixture was then warmed up to room temperature and stirred for another 3 h. The solvent was removed *in vacuo* and the residue was subjected to purification by flash column chromatography on silica gel (PE/EA: 15/1) to deliver ester **24** (315 mg, 65% yield) as colorless oil. $[\alpha]_D^{25}$ -2.4 ($c = 1.22$, $CHCl_3$). ^{1}H NMR (400 MHz, $CDCl_3$): 7.67 (dd, $J = 10.7$, 7.8, 1.4 Hz, 4H), 7.46–7.35 (m, 6H), 6.40 (dd, $J = 17.3$, 1.4 Hz, 1H), 6.13 (dd, $J = 17.3$, 10.4 Hz, 1H), 5.82 (dd, $J = 10.4$, 1.4 Hz, 1H), 5.59 (dd, $J = 17.2$, 10.2, 9.1 Hz, 1H), 5.10 (dt, $J = 5.4$, 3.1 Hz, 2H), 5.03 (dd, $J = 17.1$, 1.2 Hz, 1H), 3.85 (dd, $J = 6.7$, 5.3 Hz, 1H), 3.65 (t, $J = 9.7$ Hz, 1H), 3.56–3.46 (m, 2H), 3.40 (s, 3H), 2.26 (dq, $J = 13.8$, 4.5 Hz, 1H), 2.17–2.07 (m, 1H), 1.91–1.81 (m, 2H), 1.74–1.63 (m, 1H), 1.58–1.48 (m, 1H), 1.34–1.22 (m, 1H), 1.07 (s, 9H), 0.91 (s, 9H), 0.86 (t, $J = 7.4$ Hz, 3H), 0.79 (d, $J = 7.0$ Hz, 3H), 0.72 (d, $J = 6.9$ Hz, 3H), 0.07 (s, 3H), 0.04 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$): 165.92, 138.25, 135.73, 135.72, 134.13, 134.06, 130.43, 129.72, 129.11, 127.77, 117.19, 81.04, 75.09, 72.72, 66.53, 60.21, 49.14, 39.52, 38.17, 35.11, 27.05, 26.14, 22.29, 19.41, 18.23, 14.14, 11.87, 9.42, −4.21, −4.29. FT-IR: ν (cm^{-1}) 3072, 3050, 2930, 2857, 1724, 1638, 1463, 1428, 1389, 1259, 1194, 1112, 807, 702. HRMS-ESI Positive: calcd. for $C_{40}H_{64}O_5NaSi_2$ $[M+Na]^+$: 703.4184; found: 703.4187.

*(5R,6R)-6-((2S,3R,4R,5S)-2-((tert-butyldimethylsilyl)oxy)-6-((tert-butyldiphenylsilyl)oxy)-4-methoxy-3,5-dimethylhexyl)-5-ethyl-5,6-dihydro-2H-pyran-2-one (**25**).* To a solution of ester **24** (137 mg, 0.20 mmol) in freshly distilled CH_2Cl_2 (40 mL) at 40 °C was added a solution of Grubbs-II catalyst (34 mg, 0.04 mmol) in CH_2Cl_2 (3 mL). The reaction process was monitored by TLC. After completion, the solvent was removed under reduced pressure and the brown residue was subjected to purification by flash column chromatography on silica gel (PE/EA: 25:1) to afford pyrone **25** (105 mg, 80% yield, brsm: 91%) as colorless oil. $[\alpha]_D^{25}$ -94.3 ($c = 0.36$, $CHCl_3$). ^{1}H NMR (400 MHz, $CDCl_3$): 7.65 (t, $J = 7.8$ Hz, 4H), 7.46–7.35 (m, 6H), 7.02 (dd, $J = 9.7$, 6.1 Hz, 1H), 6.04 (d, $J = 9.7$ Hz, 1H), 4.65 (td, $J = 7.2$, 3.3 Hz, 1H), 4.08–3.99 (m, 1H), 3.63 (t, $J = 9.7$ Hz, 1H), 3.56 (dd, $J = 10.1$, 6.2 Hz, 1H), 3.44 (d, $J = 10.0$ Hz, 1H), 3.38 (s, 3H), 2.27 (td, $J = 9.7$, 4.4 Hz, 1H), 2.01–1.80 (m, 4H), 1.76–1.62 (m, 1H), 1.54–1.46 (m, 1H), 1.06 (s, 9H), 0.96 (t, $J = 7.5$ Hz, 3H), 0.88 (s, 9H), 0.82 (d, $J = 7.0$ Hz, 3H), 0.72 (d, $J = 6.9$ Hz, 3H), 0.06 (s, 3H), 0.04 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$): 164.91, 150.56, 135.59, 135.57, 133.83, 133.72, 129.69, 129.66, 127.68, 121.13, 81.22, 78.77, 70.22, 66.47, 60.19, 40.32, 38.08, 37.36, 33.68, 26.90, 25.87, 20.52, 19.25, 18.07, 12.43, 10.85, 9.33, −4.17, −4.64. FT-IR: ν (cm^{-1}) 3071, 3049, 2929, 2856, 1731, 1471, 1428, 1386, 1257, 1111, 1025, 965, 939, 902, 868, 825, 774. HRMS-ESI Positive: calcd. for $C_{38}H_{60}O_5NaSi_2$ $[M+Na]^+$: 675.3871; found: 675.3873.

To a solution of pyrone **25** (56 mg, 0.085 mmol) in freshly distilled THF (4 mL) was added dropwise tetrabutylammonium fluoride (0.17 mL, 1.0 M in THF) at ambient temperature. The resulting solution was allowed to stir for 2 h before addition of aqueous solution of NH_4Cl (1.0 mL). The organic layer was seperated and the aqueous layer was extracted with EtOAc (3 × 10 mL). The combined organic layers were washed with brine, dried over Na_2SO_4, filtered and concentrated *in vacuo*. Purification by flash chromatography on silica gel (PE/EA: 5/1) yielded the desired product **26a** (13 mg, 37% yield), an elimination product **26b** (11 mg, 20% yield), and a bridged product **26c** (8.3 mg, 32% yield) as colorless oil.

*(5R,6R)-6-((2S,3R,4R,5S)-2-((tert-butyldimethylsilyl)oxy)-6-hydroxy-4-methoxy-3,5-dimethylhexyl)-5-ethyl-5,6-dihydro-2H-pyran-2-one (**26a**).* $[\alpha]_D^{25}$ -119 ($c = 0.25$, $CHCl_3$). ^{1}H NMR (400 MHz, $CDCl_3$): 7.03 (dd, $J = 9.7$, 6.1 Hz, 1H), 6.04 (d, $J = 9.7$ Hz, 1H), 4.64 (td, $J = 7.3$, 3.3 Hz, 1H), 4.09–4.02 (m, 1H), 3.68–3.61 (m, 2H), 3.40 (s, 3H), 3.27 (dd, $J = 9.7$, 1.9 Hz, 1H), 2.28 (td, $J = 9.9$, 3.9 Hz, 1H), 2.01–1.84 (m, 4H), 1.71–1.66 (m, 1H), 1.57–1.44 (m, 1H), 0.97 (t, $J = 7.5$ Hz, 3H), 0.90–0.85 (m, 18H), 0.07 (s, 3H), 0.05 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$): 164.95, 150.66, 121.27, 83.37, 78.79, 70.19, 66.43, 59.93, 40.30, 38.02, 37.56, 33.77, 25.98, 20.67, 18.21, 12.53, 11.03, 10.46, −4.03, −4.49. FT-IR: ν (cm^{-1}) 3453, 2959, 2929, 2856, 1731, 1462, 1386, 1360, 1254, 1069, 1047, 966, 939, 868, 834, 774. HRMS-ESI Positive: calcd. for $C_{22}H_{43}O_5Si$ $[M+H]^+$: 415.2874; found: 415.2873.

*(2Z,4E,7S,8R,9R,10S)-7-((tert-butyldimethylsilyl)oxy)-11-((tert-butyldiphenylsilyl)oxy)-4-ethyl-9-methoxy-8,10-dimethylundeca-2,4-dienoic acid (**26b**).* $[\alpha]_D^{25}$ -16.7 ($c = 0.065$, $CHCl_3$). ^{1}H NMR (400 MHz, $CDCl_3$): 7.70–7.62 (m, 4H), 7.45–7.35 (m, 6H), 6.51 (d, $J = 12.6$ Hz, 1H), 5.79 (t, $J = 7.1$ Hz, 1H), 5.70 (d, $J = 12.5$ Hz, 1H), 3.99–3.93 (m, 1H), 3.64 (t, $J = 9.6$ Hz, 1H), 3.54 (dd, $J = 10.0$, 6.0 Hz, 1H), 3.41–3.36 (m, 4H), 2.43–2.31 (m, 3H), 2.28–2.18 (m, 1H), 1.89–1.79 (m, 2H), 1.06 (s, 9H), 0.99 (t, $J = 7.5$ Hz, 3H), 0.88 (s, 9H), 0.79 (d, $J = 7.0$ Hz, 3H), 0.71 (d, $J = 6.9$ Hz, 3H), 0.03 (s, 3H), 0.02 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$): 168.56, 147.94, 138.95, 135.58, 134.56, 133.89, 133.85, 129.63, 127.65, 116.67, 81.47, 73.01, 66.32, 60.59, 41.85, 37.85, 31.03, 26.91, 25.88, 22.49, 19.27, 18.05, 13.34, 11.19, 9.18, −4.49, −4.51. FT-IR: ν (cm^{-1}) 3071, 2961, 2928, 2856, 1698, 1614, 1462, 1428, 1260, 1083, 938.863, 800, 739, 701. HRMS-ESI Positive: calcd. for $C_{38}H_{60}O_5NaSi_2$ $[M+Na]^+$: 675.3871; found: 675.3874.

*(1R,5R,7S,9R)-9-ethyl-7-((2S,3R,4S)-5-hydroxy-3-methoxy-4-methylpentan-2-yl)-2,6-dioxabicyclo[3.3.1]nonan-3-one (**26c**).* $[\alpha]_D^{25}$ +27.6 ($c = 0.015$, $CHCl_3$). ^{1}H NMR (400 MHz, $CDCl_3$): 4.73–4.68 (m, 1H), 4.13 (d, $J = 4.8$ Hz, 1H), 4.00–3.91 (m, 1H), 3.58 (d, $J = 5.2$ Hz, 2H), 3.39 (s, 3H), 3.23 (dd, $J = 7.8$, 3.1 Hz, 1H), 2.97 (d, $J = 19.1$ Hz, 1H), 2.77 (dd, $J = 19.2$, 5.4 Hz, 1H), 2.10–2.00 (m, 1H), 1.92–1.79 (m, 4H), 1.78–1.65 (m, 2H), 1.58 (s, 9H), 1.01 (t, $J = 7.2$ Hz, 3H), 0.88 (d, $J = 6.9$ Hz, 6H). ^{13}C NMR (100 MHz, $CDCl_3$): 170.16, 82.40, 76.33, 67.95, 66.42, 65.84, 60.19, 40.12, 39.17, 37.29, 36.88, 29.33, 27.72, 21.12, 11.59, 11.29, 10.90. FT-IR: ν (cm^{-1}) 3360, 2962, 2923, 2852, 1733, 1468, 1359, 1261, 1221, 1181, 1084, 1031, 863, 799. HRMS-ESI Positive: calcd. for $C_{16}H_{28}O_5Na$ $[M+Na]^+$: 323.1829; found: 323.1830.

*(5R,6R)-6-((2S,3R,4R,5S)-(E)-2-((tert-butyldimethylsilyl)oxy)-4-methoxy-3,5-dimethyl-8-oxonon-6-en-1-yl)-5-ethyl-5,6-dihydro-2H-pyran-2-one (**32**).* A solution of alcohol **26a** (14.7 mg, 0.035 mmol) in CH_2Cl_2 (5 mL) was treated with Dess-Martin periodinane (60 mg, 0.14 mmol) in one portion at 0 °C. After 10 min, the solution was warmed to room temperature and stirred for 2 h. The reaction was then quenched by addition of saturated $NaHCO_3$ solution and $Na_2S_2O_3$ solution, respectively. The aqueous layer was extracted with EtOAc (3 × 10 mL). The combined organic layers were washed with brine, dried over Na_2SO_4 and concentrated. The residue was used directly for the next step without further purification. To a solution of dimethyl acetylmethylphosphonate (36 μL, 0.26 mmol) in anhydrous THF (3.0 mL) was slowly added potassium *tert*-butoxide (215 μL, 1.0 M in THF) at 0 °C. After stirring for 1 h at room temperature, a solution of the aforementioned crude product in THF (2.0 mL) was then added dropwise at 0 °C. The resulting mixture was warmed to room temperature and kept stirring for 4 h, after which the reaction was quenched by a few drops of water. The solvent was removed under reduced pressure and the residue was subjected to purification on silica gel (PE/EA: 5/1) to yield enone **32** (13.3 mg, 84%) as colorless oil. $[\alpha]_D^{25}$ -135.3 ($c = 0.12$, $CHCl_3$). ^{1}H NMR (400 MHz, $CDCl_3$): 7.02 (dd, $J = 9.7$, 6.1 Hz, 1H), 6.95 (dd, $J = 16.1$,

6.6 Hz, 1H), 6.13 (dd, $J = 16.1, 1.5$ Hz, 1H), 6.04 (d, $J = 9.8$ Hz, 1H), 4.63 (td, $J = 7.8, 3.4$ Hz, 1H), 4.08–3.99 (m, 1H), 3.29 (s, 3H), 3.14 (dd, $J = 9.3, 2.5$ Hz, 1H), 2.65–2.55 (m, 1H), 2.30–2.23 (m, 4H), 1.99–1.88 (m, 2H), 1.86–1.78 (m, 1H), 1.72–1.62 (m, 1H), 1.54–1.46 (m, 1H), 1.09 (d, $J = 6.8$ Hz, 3H), 0.97 (t, $J = 7.5$ Hz, 3H), 0.90 (d, $J = 6.9$ Hz, 3H), 0.87 (s, 9H), 0.05 (s, 3H), 0.03 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$): 198.50, 164.91, 151.89, 150.56, 130.03, 121.28, 86.50, 78.63, 59.83, 40.70, 38.82, 37.43, 33.69, 27.45, 25.96, 20.57, 18.19, 12.51, 12.26, 11.02, −4.00, −4.49. FT-IR: ν (cm^{-1}) 2929, 2856, 1731, 1676, 1623, 1463, 1386, 1360, 1255, 1089, 1068, 1026, 965, 834. HRMS-ESI Positive: calcd. for $C_{25}H_{44}O_5NaSi$ $[M+Na]^+$: 475.2850; found: 475.2852.

(3R,4R,6R,7R,8R,9S)-6-((tert-butyldimethylsilyl)oxy)-10-((tert-butyldiphenylsilyl)oxy)-3-ethyl-8-methoxy-7,9-dimethyldec-1-en-4-yl *cinnamate (**35**).* To a suspension of *cis*-2-pentene (0.45 mL, 4.23 mmol) and potassium *tert*-butoxide (1.4 mL, 1.0 M in THF) in freshly distilled THF (1.5 mL) was slowly added ^{n}BuLi (0.59 mL, 2.4 M in hexane) at −78 °C. The resulting solution was then warmed to −50 °C and kept stirring for 10 min. The reaction mixture was re-cooled to −78 °C and then added a solution of (−)-methoxydiisopinocampheylborane (558 mg, 1.76 mmol) in THF (3.5 mL). The colorless solution was stirred for 30 min at −78 °C and slowly added boron trifluoride etherate (217 μL, 1.76 mmol) and compound **14** (393 mg, 0.71 mmol), respectively. The reaction mixture was stirred for an additional 6 h until the TLC indicated a full conversion of the starting material. The reaction was quenched by addition of aqueous solution of NaOH (1.1 mL, 3.0 M) and 30 wt% H_2O_2 (1.1 mL), respectively. The corresponding mixture was heated up and refluxed for 1 h, after which the aqueous layer was extracted with EtOAc (3 × 10 mL). The combined organic layers were washed with brine, dried over Na_2SO_4, and concentrated under reduced pressure. The residue was re-dissolved in CH_2Cl_2 (20 mL) and added triethylamine (1.1 mL, 7.6 mmol), 4-Dimethylaminopyridine (310 mg, 2.54 mmol) and cinnamoyl chloride (847 mg, 5.1 mmol) at 0 °C. The reaction mixture was warmed up to 40 °C and stirred for another 5 h. After completion, the solvent was removed *in vacuo*. The residue was subjected to purification by flash column chromatography on silica gel (PE/EA: 150/1) to yield ester **35** (361 mg, 68% yield) as colorless oil. $[\alpha]_D^{25}$ +23.4 ($c = 0.48$, $CHCl_3$). ^{1}H NMR (400 MHz, $CDCl_3$): 7.68–7.63 (m, 5H), 7.54–7.48 (m, 2H), 7.43–7.34 (m, 9H), 6.42 (d, $J = 16.0$ Hz, 1H), 5.67–5.54 (m, 1H), 5.13 (dd, $J = 10.3, 1.8$ Hz, 1H), 5.08–4.98 (m, 2H), 4.11 (t, $J = 6.3$ Hz, 1H), 3.62 (t, $J = 9.6$ Hz, 1H), 3.51–3.42 (m, 5H), 2.31–2.21 (m, 1H), 1.93–1.77 (m, 3H), 1.60–1.48 (m, 2H), 1.37–1.27 (m, 1H), 1.04 (s, 9H), 0.92–0.87 (m, 13H), 0.76 (d, $J = 6.9$ Hz, 3H), 0.63 (d, $J = 6.9$ Hz, 3H), 0.10 (s, 3H), 0.09 (s, 3H). ^{13}C NMR (125 MHz, $CDCl_3$): 166.63, 144.52, 138.15, 135.72, 134.72, 134.09, 134.05, 130.27, 129.68, 129.66, 128.99, 128.18, 127.75, 127.73, 118.73, 117.65, 80.75, 74.85, 69.81, 66.89, 60.99, 50.00, 41.25, 37.89, 37.67, 27.04, 26.26, 22.88, 19.37, 18.46, 12.02, 9.81, 9.23, −3.22, −3.70. FT-IR: ν (cm^{-1}) 3070, 2959, 2929, 2857, 1712, 1638, 1471, 1462, 1450, 1427, 1387, 1360, 1326, 1308, 1259, 1201, 1171, 1023, 917, 863, 803. HRMS-ESI Positive: calcd. For $C_{46}H_{69}O_5Si_2$ $[M+H]^+$: 757.4678; found: 757.4674.

(3R,4R,6R,7R,8R,9S)-6-((tert-butyldimethylsilyl)oxy)-3-ethyl-10-hydroxy-8-methoxy-7,9-dimethyldec-1-en-4-yl *cinnamate (**36**).* To a solution of ester **35** (46.6 mg, 0.06 mmol) in freshly distilled THF (3.0 mL) was slowly added tetrabutylammonium fluoride (185 μL, 1.0 M in THF). The resulting solution was allowed to stir overnight at room temperature. After completion, the reaction was quenched by a few drops of aqueous solution of NH_4Cl. The aqueous layer was then extracted with EtOAc (3 × 10 mL). The combined organic layers were washed with brine, dried over Na_2SO_4, filtered and concentrated. The residue was subjected to purification by flash column chromatography on silica gel (PE/EA: 15/1) to afford desired alcohol **36** (25.7 mg, 82%) as colorless oil. $[\alpha]_D^{25}$ +20.4 ($c = 0.26$, $CHCl_3$). ^{1}H NMR (400 MHz, $CDCl_3$): 7.67 (d, $J = 16.0$ Hz, 1H), 7.56–7.48 (m, 2H), 7.43–7.35 (m, 3H), 6.43 (d, $J = 16.0$ Hz, 1H), 5.60 (dt, $J = 17.1, 9.9$ Hz, 1H), 5.14 (dd, $J = 10.3, 1.7$ Hz, 1H), 5.09–4.98 (m, 2H), 4.07 (t, $J = 6.0$ Hz, 1H), 3.67–3.56 (m, 2H), 3.41 (s, 3H), 3.31 (dd, $J = 9.7, 1.7$ Hz, 1H), 2.32–2.21 (m, 1H), 1.93–1.76 (m, 3H), 1.68–1.61 (m, 1H), 1.59–1.48 (m, 1H), 1.37–1.27 (m, 1H), 0.95–0.86 (m, 12H), 0.81 (d, $J = 7.0$ Hz, 3H), 0.78 (d, $J = 6.9$ Hz, 3H), 0.09 (s, 3H), 0.08 (s, 3H). ^{13}C NMR (150 MHz, $CDCl_3$): 166.68, 144.66, 138.07, 134.69, 130.33, 129.01, 128.20, 118.64, 117.72, 82.81, 74.80, 69.84, 66.89, 60.38, 50.09, 41.47, 37.64, 26.20, 26.18, 22.95, 18.50, 12.01, 10.02, 9.82, −3.28, −3.87. FT-IR: ν (cm^{-1}) 2958, 2928, 2855, 1712, 1638, 1462, 1450, 1380, 1359, 1309, 1254, 1202, 1172, 1082, 1041, 979, 918, 864, 834, 769. HRMS-ESI Positive: calcd. For $C_{30}H_{50}O_5NaSi$ $[M+Na]^+$: 541.3320; found: 541.3317.

(3R,4R,6R,7R,8R,9S)-(E)-6-((tert-butyldimethylsilyl)oxy)-3-ethyl-8-methoxy-7,9-dimethyl-12-oxotrideca-1,10-dien-4-yl *cinnamate (**37**).* A solution of alcohol **36** (19.4 mg, 0.037 mmol) in anhydrous CH_2Cl_2 (3 mL) was treated with Dess-Martin periodinane (60 mg, 0.14 mmol) at 0 °C. After 10 min, the resulting solution was warmed up to room temperature and stirred for 2 h. After completion, the reaction was quenched by addition of saturated aqueous $NaHCO_3$ solution and $Na_2S_2O_3$ solution, respectively. The aqueous layer was extracted with EtOAc (3 × 10 mL). The combined organic layers were washed with brine, dried over Na_2SO_4 and concentrated. The residue was used directly for next step without further purification. To a solution of dimethyl acetylmethylphosphonate (38 μL, 0.27 mmol) in anhydrous THF (2.0 mL) was slowly added potassium *tert*-butoxide (227 μL, 1.0 M in THF) at 0 °C. After stirring for 1 h at room temperature, a solution of the aforementioned crude product in anhydrous THF (2 mL) was added dropwise at 0 °C. The resulting solution was then warmed to room temperature and kept stirring overnight, after which the reaction was quenched by a few drops of water. The solvent was removed under reduced pressure and further purification on silica gel (PE/EA: 10/1) yielded enone **37** (15.8 mg, 78%) as colorless oil. $[\alpha]_D^{25}$ +16.8 (c = 0.26, $CHCl_3$). ^{1}H NMR (400 MHz, $CDCl_3$) 7.66 (d, $J = 16.0$ Hz, 1H), 7.55–7.49 (m, 2H), 7.42–7.35 (m, 3H), 6.94 (dd, $J = 16.1, 6.6$ Hz, 1H), 6.42 (d, $J = 16.0$ Hz, 1H), 6.07 (dd, $J = 16.1, 1.3$ Hz, 1H), 5.65–5.53 (m, 1H), 5.14 (dd, $J = 10.3, 1.7$ Hz, 1H), 5.05 (dd, $J = 17.1, 1.4$ Hz, 1H), 5.03–4.94 (m, 1H), 4.08 (t, $J = 6.2$ Hz, 1H), 3.31 (s, 3H), 3.13 (dd, $J = 12.7, 6.3$ Hz, 1H), 2.59–2.49 (m, 1H), 2.29–2.19 (m, 4H), 1.92–1.74 (m, 2H), 1.60–1.46 (m, 2H), 1.37–1.26 (m, 1H), 0.99 (d, $J = 6.8$ Hz, 3H), 0.93–0.85 (m, 12H), 0.80 (d, $J = 6.9$ Hz, 3H), 0.09 (s, 3H), 0.06 (s, 3H). ^{13}C NMR (151 MHz, $CDCl_3$): 198.79, 166.65 152.91, 144.70, 138.04, 134.66, 130.37, 129.77, 129.03, 128.20, 118.58, 117.76, 85.45, 74.69, 69.53, 60.39, 50.16, 41.69, 38.52, 37.70, 27.27, 26.20, 22.93, 18.47, 11.99, 11.57, 9.72–3.31, −4.04. FT-IR: ν (cm^{-1}) 3357, 3187, 3062, 2927, 2854, 1711, 1677, 1637, 1578, 1496, 1450, 1359, 1308, 1254, 1202, 1172, 1094, 1024, 918, 864, 835, 805, 769. HRMS-ESI Positive: calcd. For $C_{33}H_{52}O_5NaSi$ $[M+Na]^+$: 579.3476; found: 579.3487.

(5R,6R)-6-((2R,3R,4R,5S)-(E)-2-((tert-butyldimethylsilyl)oxy)-4-methoxy-3,5-dimethyl-8-oxonon-6-en-1-yl*)-5-ethyl-5,6-dihydro-2H-pyran-2-one (**30**).* A solution of enone **37** (15.8 mg, 0.028 mmol) in freshly distilled toluene (5 mL) was warmed up to 100 °C, followed by slow addition of Grubbs-II catalyst (4.8 mg, 0.0057 mmol) in toluene (0.5 mL). The reaction process was monitored by TLC. After completion, the solvent was removed under reduced pressure. Further purification of the brown residue by flash column chromatography on silica gel (PE/EA: 5:1) afforded pyrone **30** (10.1 mg, 80% yield) as colorless oil. $[\alpha]_D^{25}$ -67.5 ($c = 0.87$, $CHCl_3$). ^{1}H NMR (400 MHz, $CDCl_3$): 7.01 (dd, $J = 9.8$, 6.0 Hz, 1H), 6.96 (dd, $J = 16.1, 6.7$ Hz, 1H), 6.12 (dd, $J = 16.1, 1.3$ Hz, 1H), 6.02 (d, $J = 10.0$ Hz, 1H), 4.54–4.46 (m, 1H), 4.22 (t, $J = 6.3$ Hz, 1H), 3.36 (s, 3H), 3.20 (dd,

$J = 9.2$, 2.2 Hz, 1H), 2.65–2.55 (m, 1H), 2.27 (s, 3H), 2.24–2.17 (m, 1H), 1.94 (dd, $J = 15.0$, 9.1, 6.2 Hz, 1H), 1.75–1.65 (m, 2H), 1.62–1.48 (m, 2H), 1.06 (d, $J = 6.8$ Hz, 3H), 0.97 (t, $J = 7.5$ Hz, 3H), 0.88 (s, 9H), 0.83 (d, $J = 7.0$ Hz, 3H), 0.10 (s, 3H), 0.07 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$): 198.72, 164.47, 152.43, 150.70, 129.86, 120.98, 85.57, 77.38, 68.22, 60.30, 41.72, 38.62, 38.50, 37.68, 27.40, 26.15, 20.93, 18.52, 11.96, 11.10, 10.31, −3.61, −3.70. FT-IR: ν (cm^{-1}) 2958, 2930, 2856, 1727, 1677, 1625, 1462, 1387, 1360, 1255, 1095, 1069, 834. HRMS-ESI Positive: calcd. for $C_{25}H_{44}O_5NaSi$ $[M+Na]^+$: 475.2850; found: 475.2848.

(5R,6R)-5-ethyl-6-((2R,3S,4R,5S)-(E)-2-hydroxy-4-methoxy-3,5-dimethylnon-7-en-1-yl*)-5,6-dihydro-2H-pyran-2-one (**1**).* To a solution of pyrone **30** (8.0 mg, 0.017 mmol) in MeOH (2.0 mL) was added 4-methylbenzenesulfonhydrazide (6.6 mg, 0.035 mmol) in one portion. The reaction mixture was then warmed to 60 °C until the complete consumption of the starting material, after which the solvent was removed under reduced pressure. A solution of sodium borohydride (14 mg, 0.40 mmol) in glacial acetic acid (2 mL) was added dropwise. The reaction was monitored by TLC. After being stirred at room temperature for 3 h, the reaction mixture was treated with addition of a concentrated aqueous solution of HCl to adjust the solution to pH 2–3. The resulting solution was then stirred at 70 °C for further 3 h and then slowly added an aqueous solution of $NaHCO_3$ (2 N, 5 mL) at 0 °C. The aqueous layer was extracted with EtOAc (3 × 10 mL). The combined organic layers were washed with brine, dried over Na_2SO_4, and concentrated. The residue was subjected to purification by flash column chromatography on silica gel (PE/EA: 3/1) to deliver the target molecule (−)-pironetin (**1**) (4.2 mg, 76% yield) as white solid. $[\alpha]_D^{25}$ -134.7 ($c = 0.06$, $CHCl_3$), lit [8a]:$[\alpha]_D^{20}$ -136.6 ($c = 1.0$, $CHCl_3$). ^{1}H NMR (500 MHz, $CDCl_3$): 7.01 (dd, $J = 9.8$, 6.0 Hz, 1H), 6.03 (d, $J = 9.7$ Hz, 1H), 5.48–5.33 (m, 2H), 4.74 (dt, $J = 9.1$, 3.6 Hz, 1H), 4.21 (brd, $J = 9.0$ Hz, 1H), 3.47 (s, 3H), 3.44 (brs, 1H), 2.98 (dd, $J = 6.3$, 4.3 Hz, 1H), 2.33–2.25 (m, 1H), 2.13–2.06 (m, 1H), 1.92–1.82 (m, 2H), 1.81–1.75 (m, 1H), 1.73–1.65 (m, 6H), 1.58–1.46 (m, 1H), 1.02–0.95 (m, 9H). ^{13}C NMR (100 MHz, $CDCl_3$): 164.90, 150.86, 128.89, 127.09, 120.97, 91.32, 77.90, 67.55, 61.76, 39.26, 39.06, 37.41, 36.85, 36.29, 20.93, 18.14, 15.42, 12.37, 11.17. FT-IR: ν (cm^{-1}) 2965, 2931, 1720, 1460, 1381, 1251, 1079, 1027, 966, 822. HRMS-ESI Positive: calcd. for $C_{19}H_{33}O_4$ $[M+H]^+$: 325.2373; found: 325.2374.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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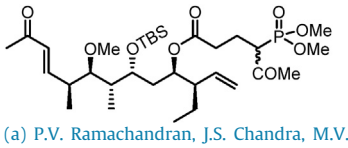
Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.tet.2020.131660.

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